

Denial of Service Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacks and Exploits

Objective: 4.2: Given a scenario, perform network attacks using the appropriate tools.

Video Overview

This video provides a foundational understanding of Denial of Service (DoS) attacks, exploring various types from basic DoS to more sophisticated application-layer attacks. As a penetration tester, comprehending these attack methodologies is essential for stress-testing systems and enhancing their resilience against real-world cyber threats. The concepts discussed here are directly applicable to securing infrastructure and operational technology environments, where maintaining service availability is very important. Understanding these attack vectors will enable you to identify vulnerabilities and help organizations develop robust defense mechanisms, ultimately contributing to a more secure digital landscape.

Key Topics Covered

- **Denial of Service (DoS) and Distributed Denial of Service (DDoS) fundamentals**
- **Reflection attacks:** attacker spoofs target IP and sends requests to a third-party server (reflector)
- **Amplification attacks:** small query sent to reflector results in a much larger payload to the target
- **Volumetric attacks:** saturating network bandwidth and overwhelming a service
- **Tools for volumetric attacks:** hping3
- **Specialized volumetric attacks:** Smurf and Fraggle
- **Application-layer attacks:** targeting HTTP or HTTPS protocols
- **Specific application attack:** Slowloris

Student Notes Section